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Nelder - Mead Method OR Simplex Search Method

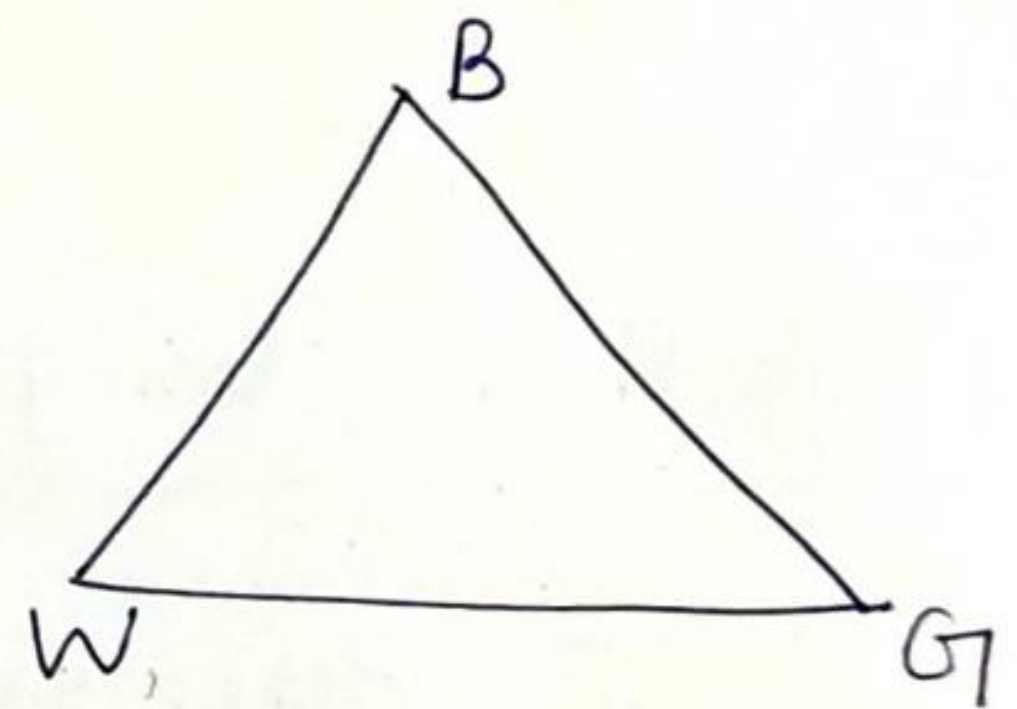
The Nelder Mead technique was proposed by "John Nelder" and "Roger Mead" in 1965.

This method is used for minimization problems only.

Nelder - Mead method is also known as simplex search method.

A simplex is a triangle that compares the function values at the three vertices of a triangle

The worst vertex W is largest, it is rejected



and replaced with new vertex. A new triangle is formed and search is continued. This process generates a sequence of triangle for which the function values get smaller and smaller. The size of the triangle is reduced and coordinates of the minimum points are found.

This method is followed by the following steps of construction as below;

Initial Triangle BGW:-

Given $f(x, y)$ and function is only used for minimum.

We have to use three vertices of triangle as

$$B = (x_1, y_1), \quad G = (x_2, y_2), \quad W = (x_3, y_3)$$

Where
B is Best vertex,
G is Good vertex.
W is Worst vertex.

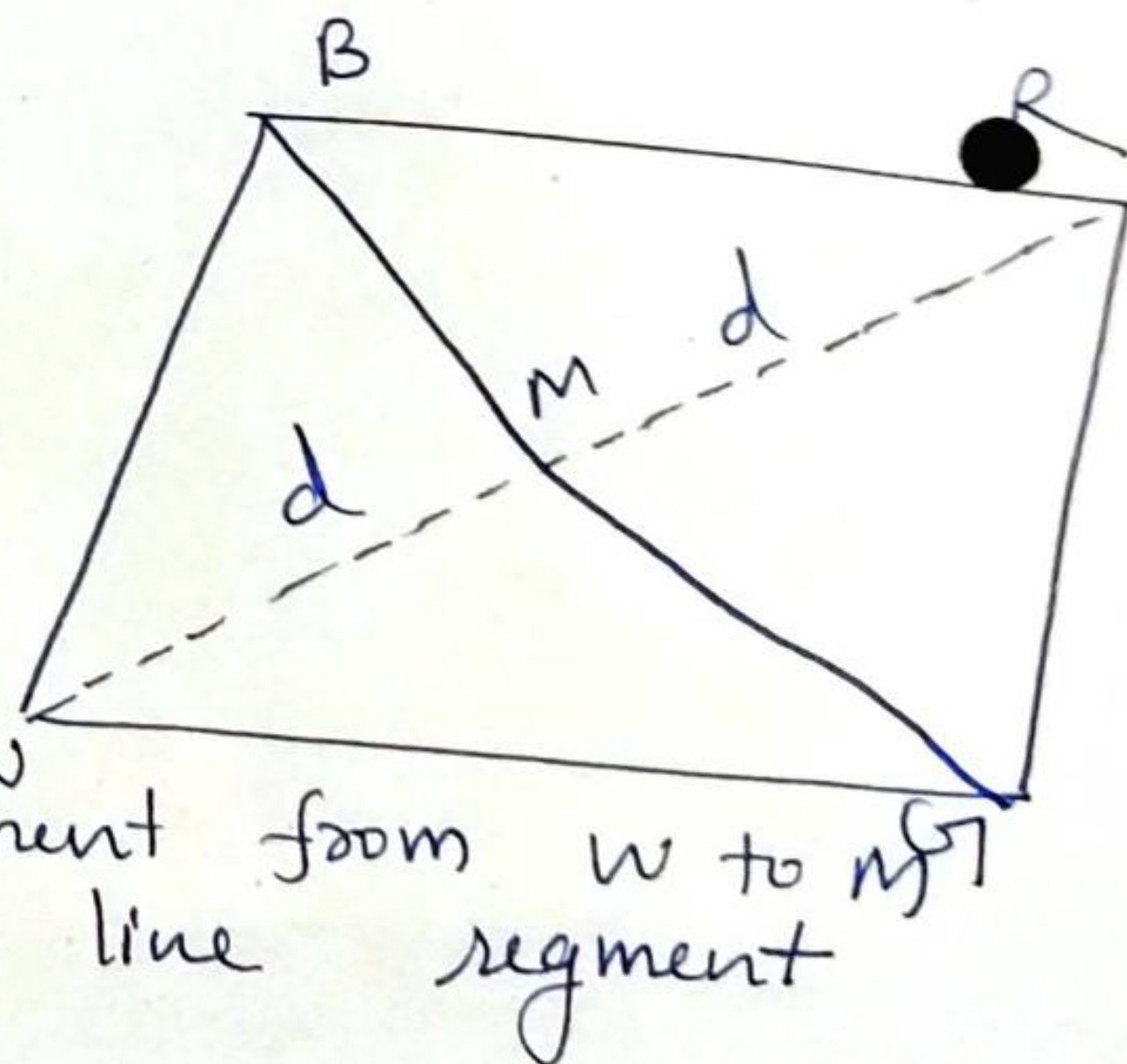
Midpoint of the Good side:-

$$M = \frac{B+G}{2} = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2} \right)$$

Reflection using point R:-

To determine R;
First of all, we find
mid point M of the
side BG.

Then we draw the line segment from W to M
as d and also extend the line segment
from M to R as d

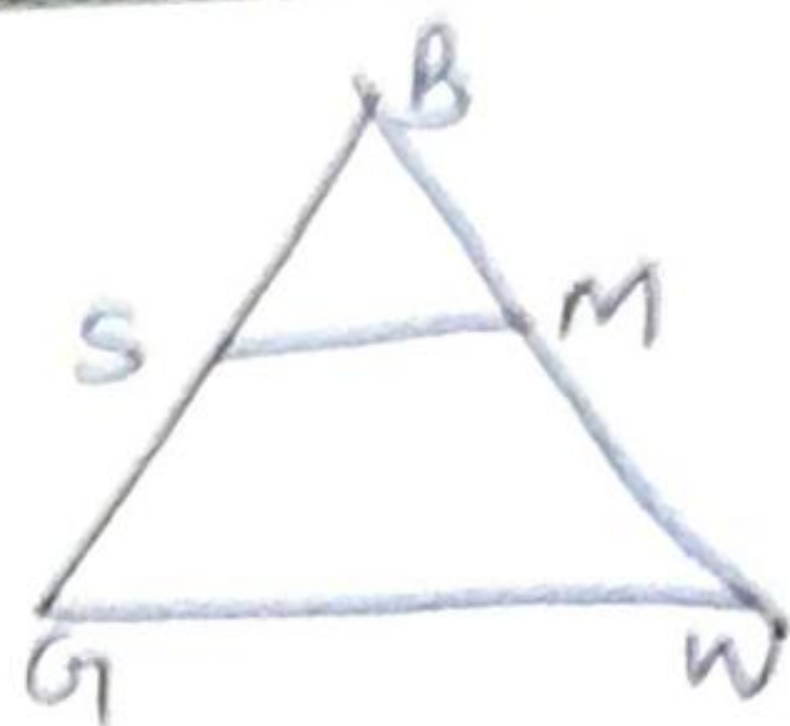


Then formula of R can be written as

$$R = M + (M - W) = 2M - W$$

Shrink toward B:-

If the functional value at C
is not less than the
functional value at W
i.e. $f(C) \geq f(W)$



Then point G and W must be shrunk
toward B.

The point G is replaced with M and W
is replaced with S.

If $f(R) < f(G)$

Perform Case (i)

(then we have to
either Reflection or Extension)

If $f(R) \geq f(G)$

Perform Case (ii')

(then we have to either
Contraction or Shrink)

Case (i)

If $f(B) < f(R)$, then
replace W with R.
Compute E and $f(E)$.
If $f(E) < f(B)$, then
replace W with E.
Else

Replace W with R
Case (i) end

Case (ii')

If $f(R) \leq f(W)$,
Then replace W with R
Compute $C_1 = \frac{W+M}{2}$
or $C_2 = \frac{M+R}{2}$
and $f(C) = ?$

If $f(C) < f(W)$, Then replace
W with C.

Else

Compute S and $f(S)$.
Replace W with S.
Replace G with M
Finally Case (ii) End.

Q:- Use Nelder-Mead method to find minimum of

$$f(x, y) = x^2 - 4x + y^2 - y - xy$$

Given vertices are $V_1 = (0, 0)$, $V_2 = (1.2, 0)$
and $V_3 = (0, 0.8)$

Sol:-

Iteration 1:-

First find BGW triangle

$$f(V_1) = f(0, 0) = 0$$

$$f(V_2) = f(1.2, 0) = (1.2)^2 - 4(1.2) + (0)^2 - (0) - (1.2)(0) = -3.36$$

$$f(V_3) = f(0, 0.8) = (0)^2 - 4(0) + (0.8)^2 - (0.8) - (0)(0.8) = -0.16$$

$$\text{Best } B = V_2 = (1.2, 0)$$

$$\text{Good } = G = V_3 = (0, 0.8)$$

$$\text{Worst } = W = V_1 = (0, 0)$$

Second, we find mid point of B and G

$$M = \frac{B+G}{2} = \left(\frac{1.2+0}{2}, \frac{0+0.8}{2} \right)$$

$$M = (0.6, 0.4)$$

$$\text{Now, } R = 2M - W = 2(0.6, 0.4) - (0, 0)$$

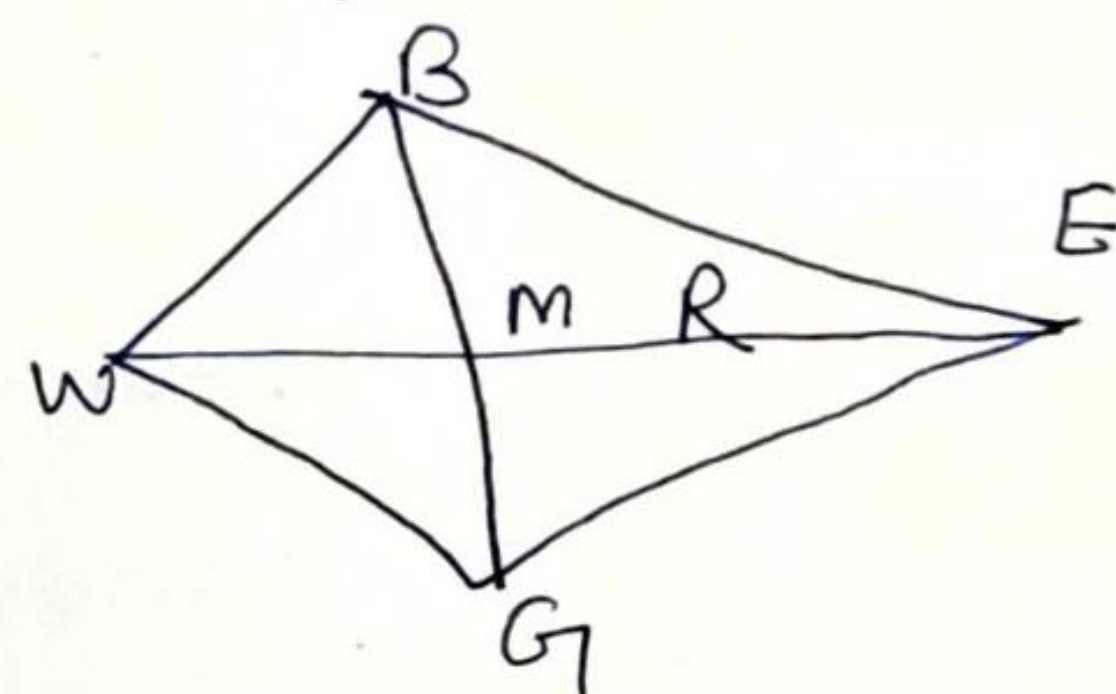
$$R = (1.2, 0.8)$$

$$f(R) = (1.2)^2 - 4(1.2) + (0.8)^2 - (0.8) - (1.2)(0.8) = -4.48$$

$$\Rightarrow f(R) < f(G) \quad \text{i.e.} \quad -4.48 < -0.16$$

Begins case (i)

$$\Rightarrow f(B) \neq f(R) \quad \text{i.e.} \quad -3.36 \neq -4.48$$



Congratulations on
earning a Ph.D
degree

Compute E and $f(E)$

$$E = 2R - M$$

$$E = 2(1.2, 0.8) - (0.6, 0.4)$$

$$E = (1.8, 1.2)$$

$$f(E) = (1.8)^2 - 4(1.8) + (1.8)^2 - (1.2) - (1.8)(1.2) \\ = -5.88$$

$\Rightarrow f(E) < f(B)$ i.e. $-5.88 < -3.36$
Replace W with E .

Iteration II:-

Now vertices are $(1.2, 0)$, $(0, 0.8)$, $(1.8, 1.2)$

$$f(1.2, 0) = -3.36$$

$$f(0, 0.8) = -0.16$$

$$f(1.8, 1.2) = -5.88$$

$$\text{Best} = B = V_2 = (1.8, 1.2)$$

$$\text{Good} = G = V_3 = (1.2, 0)$$

$$\text{Worst} = W = V_1 = (0, 0.8)$$

Midpoint of Good and Best

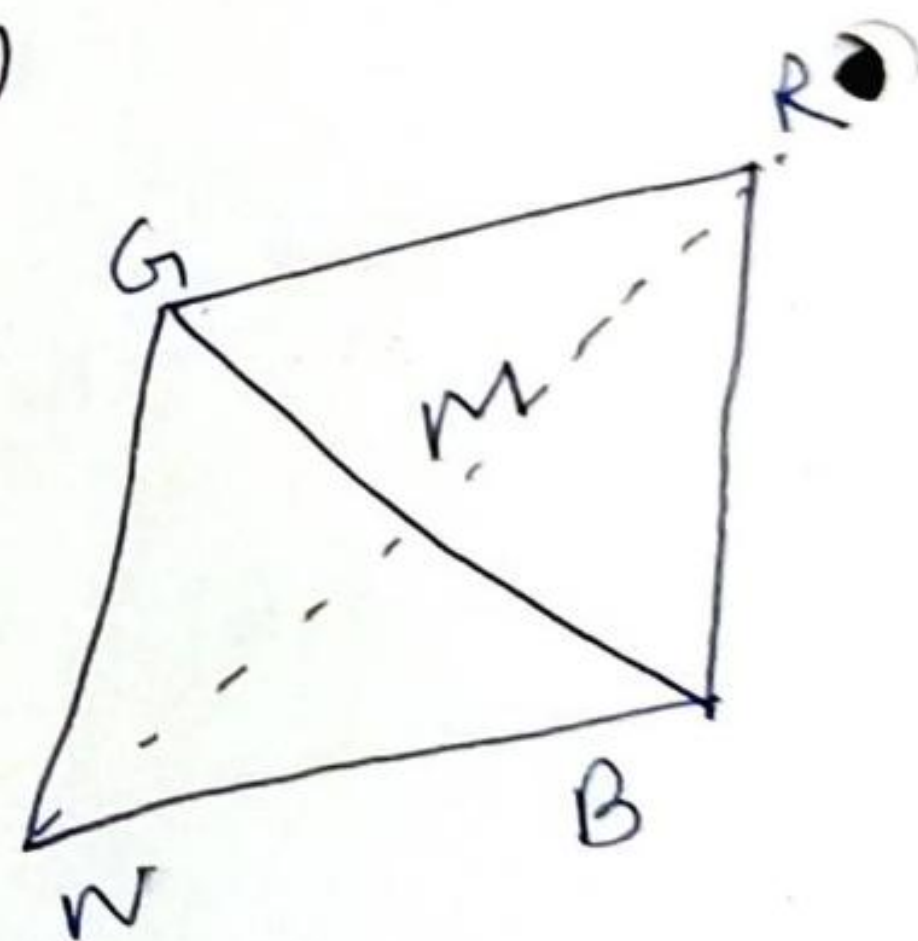
$$M = \frac{B+G}{2} = \left(\frac{1.8+1.2}{2}, \frac{1.2+0}{2} \right)$$

$$M = (1.5, 0.6)$$

$$\text{Now, } R = 2M - W$$

$$= 2(1.5, 0.6) - (0, 0.8)$$

$$R = (3, 0.4)$$



$$f(R) = (3)^2 - 4(3) + (0.4)^2 - (0.4) - (3)(0.4) \\ = -4.44 \quad (2)$$

$$\Rightarrow f(R) < f(G) \text{ i.e. } -4.44 < -3.36$$

Begins case (i)

$$\Rightarrow f(B) < f(R) \text{ i.e. } -5.88 < -4.44$$

Replace W with R

Iteration III:- find BGW triangle

Now Vertices are $(1.2, 0)$, $(1.8, 1.2)$, $(3, 0.4)$

$$f(1.2, 0) = -3.36$$

$$f(1.8, 1.2) = -5.88$$

$$f(3, 0.4) = -4.44$$

$$\text{Best} = B = V_2 = (1.8, 1.2)$$

$$\text{Good} = G = V_3 = (3, 0.4)$$

$$\text{Worst} = W = V_1 = (1.2, 0)$$

Midpoint of Good and Best

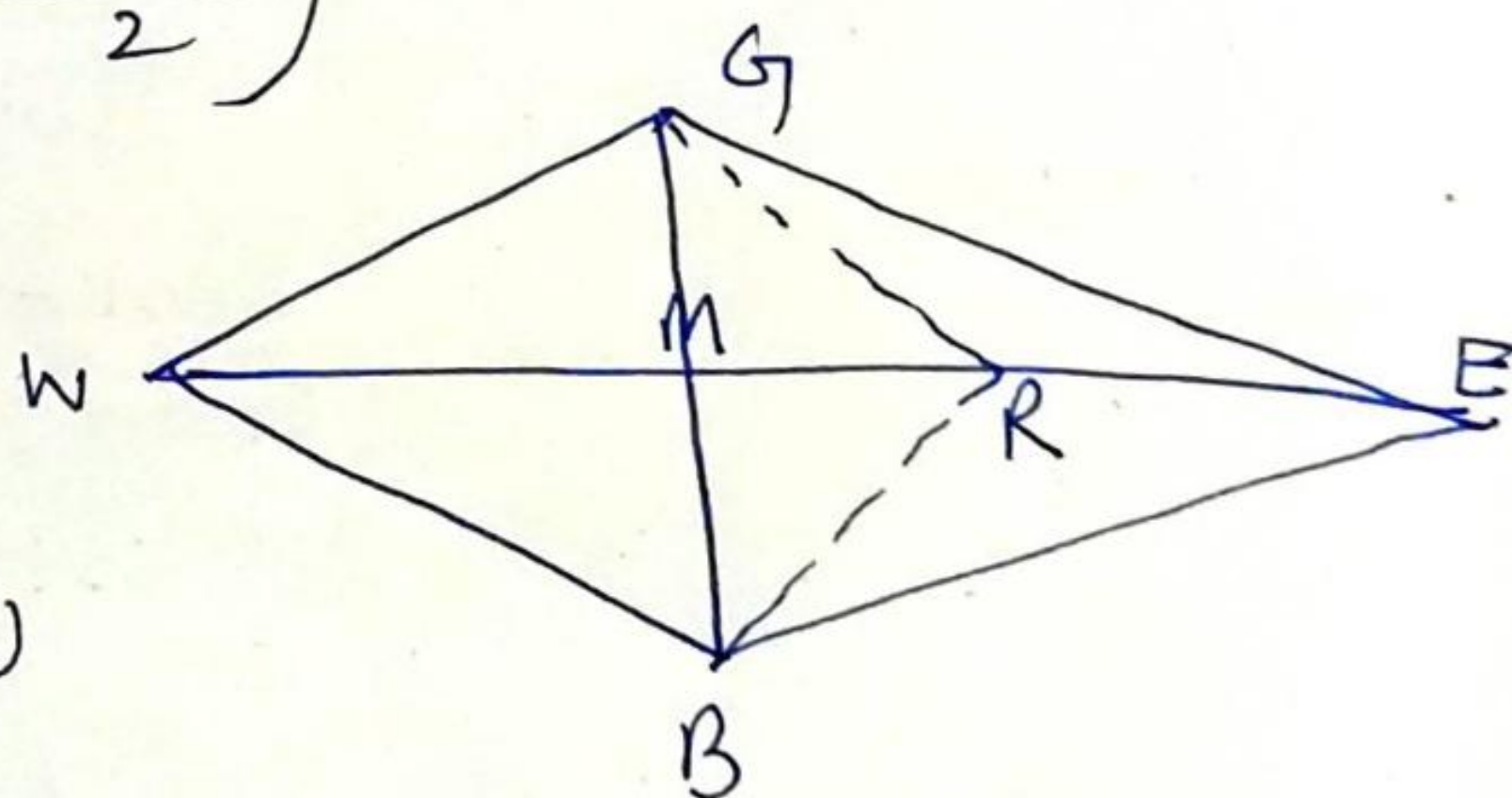
$$M = \frac{B+G}{2} = \left(\frac{1.8+3}{2}, \frac{1.2+0.4}{2} \right)$$

$$M = (2.4, 0.8)$$

$$\text{Now, } R = 2M - W$$

$$R = 2(2.4, 0.8) - (1.2, 0)$$

$$R = (3.6, 1.6)$$



$$f(R) = (3.6)^2 - 4(3.6) + (1.6)^2 - (1.6) - (3.6)(1.6) \\ = -6.24$$

$$\Rightarrow f(R) < f(G) \Rightarrow -6.24 < -4.44$$

Begin Case (i)

$$\Rightarrow f(B) < f(R) \text{ i.e. } -5.88 < -6.24$$

Compute E and f(E)

$$E = 2R - M = 2(3.6, 1.6) - (2.4, 0.8) \\ E = (4.8, 2.4)$$

$$f(E) = (4.8)^2 - 4(4.8) + (2.4)^2 - (2.4) - (4.8)(2.4) = -4.32$$

$$\Rightarrow f(E) < f(B) \text{ i.e. } -4.32 < -5.88$$

Replace W with R.

Iteration IV:-

find BGR triangle

Now vertices are $(1.8, 1.2)$, $(3, 0.4)$, $(3.6, 1.6)$

$$f(1.8, 1.2) = -5.88$$

$$f(3, 0.4) = -4.44$$

$$f(3.6, 1.6) = -6.24$$

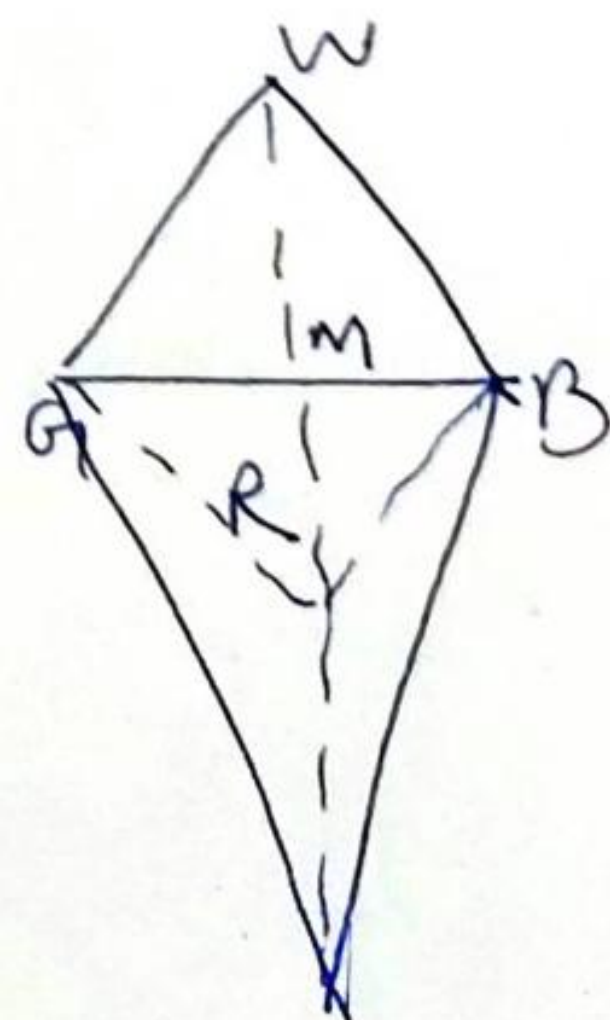
$$\text{Best} = B = V_2 = (3.6, 1.6)$$

$$\text{Good} = G = V_3 = (1.8, 1.2)$$

$$\text{Worst} = W = V_1 = (3, 0.4)$$

Mid point of Good and Best

$$M = \frac{B+G}{2} = \left(\frac{3.6+1.8}{2}, \frac{1.6+1.2}{2} \right)$$



$$M = (2.7, 1.4)$$

$$\text{Now, } R = 2M - W = 2(2.7, 1.4) - (3, 0.4)$$

$$R = (2.4, 2.4)$$

$$\begin{aligned} f(R) &= (2.4)^2 - 4(2.4) + (2.4)^2 - (2.4) - (2.4)(2.4) \\ &= -6.24 \end{aligned}$$

$$\Rightarrow f(R) < f(G) \quad \text{i.e.} \quad -6.24 < -5.88$$

Begins case (i)

$$\Rightarrow f(R) \neq f(R) \quad \text{i.e.} \quad -6.24 \neq -6.24$$

Compute E and f(E)

$$E = 2R - M = 2(2.4, 2.4) - (2.7, 1.4)$$

$$E = (2.1, 3.4)$$

$$\begin{aligned} f(E) &= (2.1)^2 - 4(2.1) + (3.4)^2 - (3.4) \\ &\quad - (2.1)(3.4) = -2.97 \end{aligned}$$

$$\Rightarrow f(E) \neq f(B) \quad \text{i.e.} \quad -2.97 \neq -6.24$$

Replace W with R.

Iteration V:- find BGW triangle

Now vertices are $(1.8, 1.2)$, $(3.6, 1.6)$, $(2.4, 2.4)$

$$f(1.8, 1.2) = -5.88$$

$$f(3.6, 1.6) = -6.24$$

$$f(2.4, 2.4) = -6.24$$

$$\text{Best} = B = (3.6, 1.6)$$

$$\text{Good} = G = (2.4, 2.4)$$

$$\text{Worst} = W = (1.8, 1.2)$$

Mid point of Good and Best

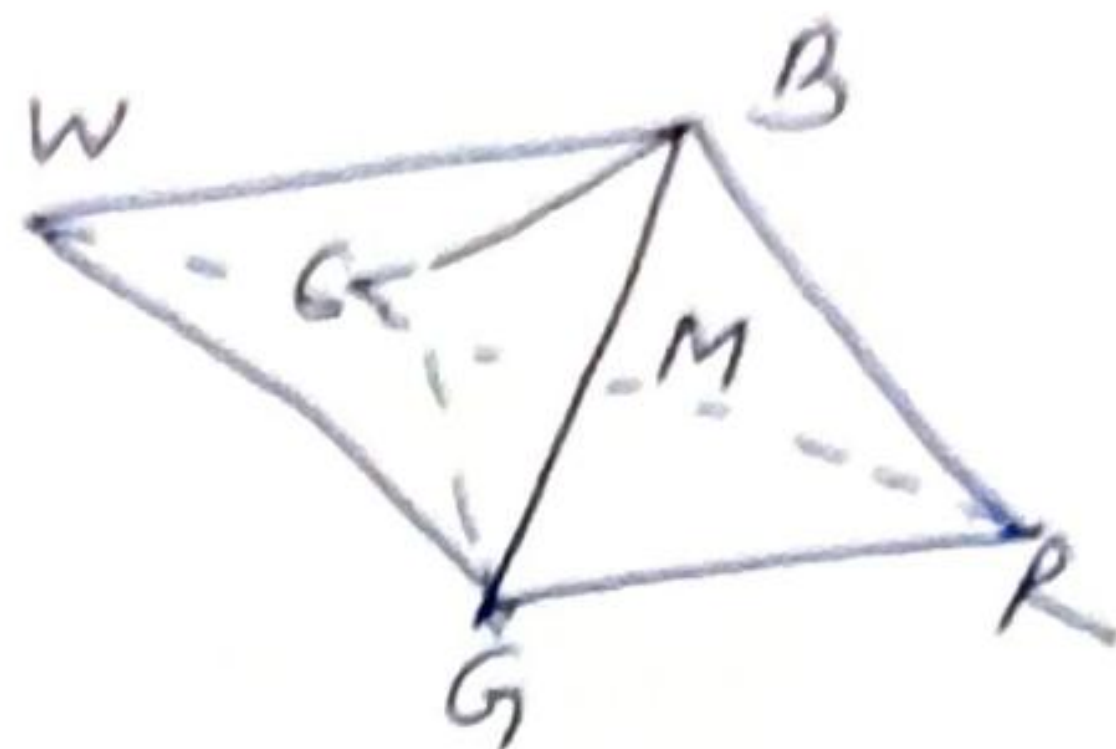
$$M = \frac{B+G}{2} = \left(\frac{3.6+2.4}{2}, \frac{1.6+2.4}{2} \right)$$

$$M = (3, 2)$$

Now $R = 2M - W$

$$R = 2(3, 2) - (1.8, 1.2)$$

$$R = (4.2, 2.8)$$



$$f(R) = (4.2)^2 - 4(4.2) + (2.8)^2 - (2.8) - (4.2)(2.8) = -5.88$$

$$f(R) \neq f(G) \quad \text{i.e.} \quad -5.88 < -6.24$$

Begins case (ii)

$$\Rightarrow f(R) \leq f(W) \quad \text{i.e.} \quad -5.88 \leq -5.88$$

Compute C

$$C_1 = \frac{W+M}{2} = \left(\frac{1.8+3}{2}, \frac{1.2+2}{2} \right) = (2.4, 1.6)$$

$$C_2 = \frac{M+R}{2} = \left(\frac{3+4.2}{2}, \frac{2+2.8}{2} \right) = (3.6, 2.4)$$

$$f(C_1) = (2.4)^2 - 4(2.4) + (1.6)^2 - (1.6) - (2.4)(1.6) = -6.72$$

$$f(C_2) = (3.6)^2 - 4(3.6) + (2.4)^2 - (2.4) - (3.6)(2.4) = -6.72$$

Both have same value

$$\text{So, } C = (2.4, 1.6)$$

$$\Rightarrow f(C) = -6.72$$

$$\Rightarrow f(c) < f(w) \text{ i.e. } -6.72 < -5.88$$

Replace w with c

Iteration VI:- Find BGV triangle

Now vertices are $(3.6, 1.6)$, $(2.4, 2.4)$, $(2.4, 1.6)$

$$f(3.6, 1.6) = -6.24$$

$$f(2.4, 2.4) = -6.24$$

$$f(2.4, 1.6) = -6.72$$

$$\text{Best} = B = (2.4, 1.6)$$

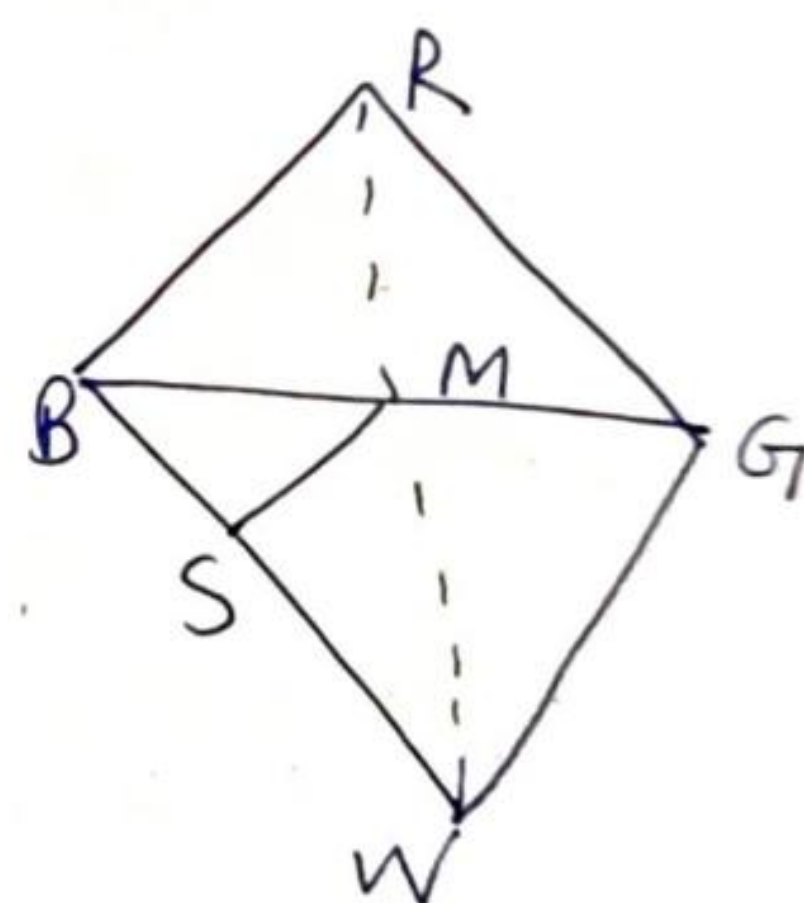
$$\text{Good} = G = (3.6, 1.6)$$

$$\text{Worst} = W = (2.4, 2.4)$$

Midpoint of Good and Best

$$M = \frac{B+G}{2} = \left(\frac{2.4+3.6}{2}, \frac{1.6+1.6}{2} \right)$$

$$M = (3, 1.6)$$



$$\text{Now, } R = 2M - W = 2(3, 1.6) - (2.4, 2.4)$$

$$R = (3.6, 0.8)$$

$$f(R) = (3.6)^2 - 4(3.6) + (0.8)^2 - (0.8) - (3.6)(0.8) = -4.48$$

$$\Rightarrow f(R) \neq f(G) \text{ i.e. } -4.48 \neq -6.24$$

Begins case (ii)

$$\Rightarrow f(R) \leq f(w)$$

$$\text{i.e. } -4.48 \leq -6.24$$

Compute S and $f(S)$ where S is the mid point of B and w .

$$S = \frac{B+w}{2} = \left(\frac{2.4+2.4}{2}, \frac{1.6+2.4}{2} \right) = (2.4, 2)$$

$$f(S) = (2.4)^2 - 4(2.4) + (2)^2 - (2) - (2.4)(2) \\ = -6.64$$

Replace w with S and replace G with M and so on.

Qo:- Use Nelder-Mead method to find minimum of

$$\min f(x, y) = |8x - y^3 + 1| + x^2 + \frac{1}{10}y^4$$

up to three iteration.

Given vertices are $V_1 = (1.5, 0)$, $V_2 = (2, 0)$ and $V_3 = (2, 0.5)$

Sol:- Iteration 1:-

First find BGW triangle

$$f(u_1) = f(1.5, 0) = 3.276$$

$$f(u_2) = f(2, 0) = 5.035$$

$$f(u_3) = f(2, 0.5) = 4.916$$

$$\text{Best} = B = (1.5, 0)$$

$$\text{Good} = G = (2, 0.5)$$

$$\text{Worst} = w = (2, 0)$$

Second we find mid point of B and G

$$M = \frac{B+G}{2} = \left(\frac{1.5+2}{2}, \frac{0+0.5}{2} \right)$$

$$M = (1.75, 0.25)$$

$$\text{Now } R = 2M - W$$

$$= 2(1.75, 0.25) - (2, 0)$$

$$R = (1.5, 0.5)$$

$$f(R) = 3.15$$

$$\Rightarrow f(R) < f(G) \quad \text{i.e.} \quad 3.15 < 4.916$$

Begins case (i)

$$\Rightarrow f(B) \neq f(R) \quad \text{i.e.} \quad 3.276 \neq 3.15$$

Compute E and $f(E)$

$$E = 2R - M = 2(1.5, 0.5) - (1.75, 0.25)$$

$$E = (1.25, 0.75)$$

$$f(E) = 2.194$$

$$\Rightarrow f(E) < f(B) \quad \text{i.e.} \quad 2.194 < 3.276$$

Replace W with E.

Iteration II:- find BGW triangle

Now vertices are $(1.5, 0)$, $(2, 0.5)$, $(1.25, 0.75)$

$$f(1.5, 0) = 3.276$$

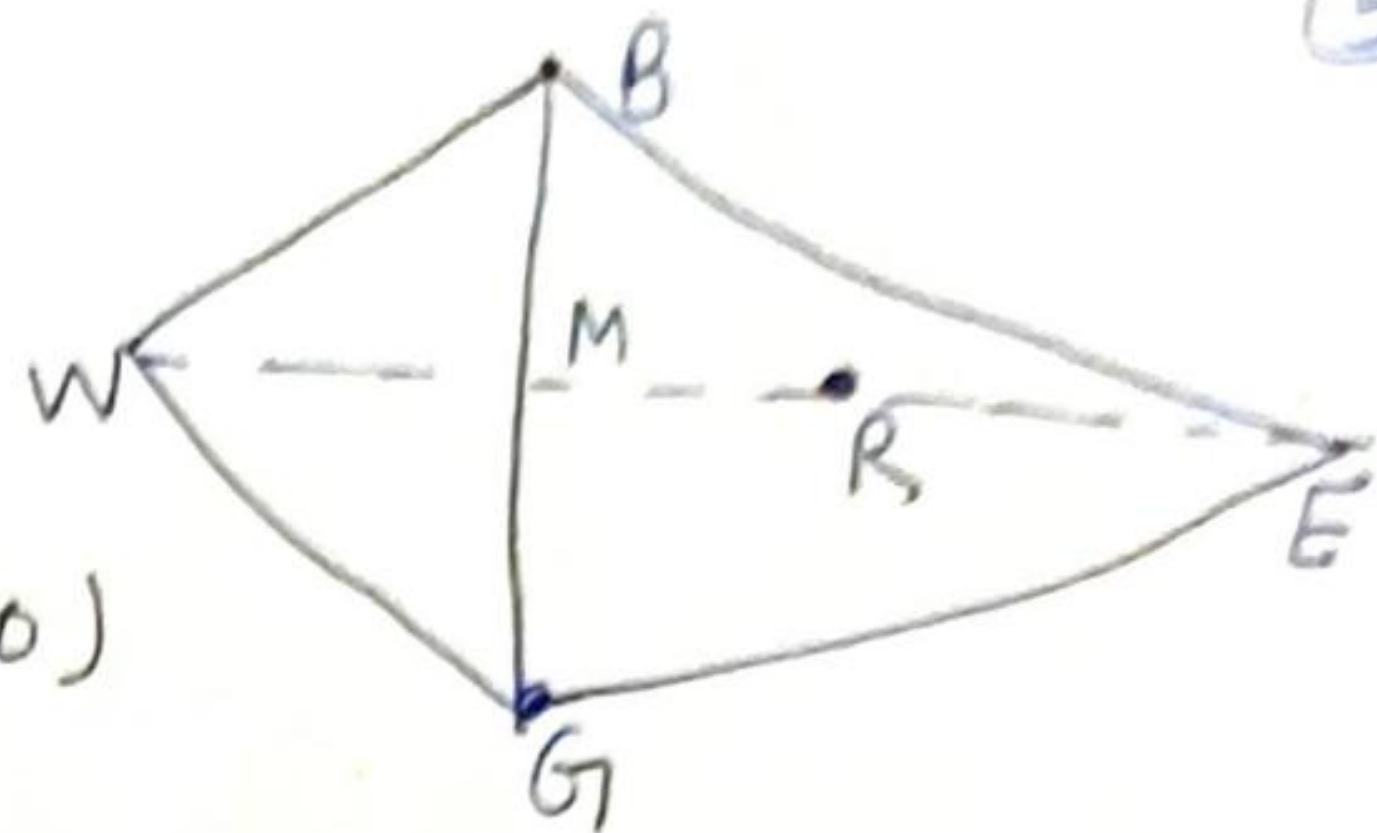
$$f(2, 0.5) = 4.916$$

$$f(1.25, 0.75) = 2.194$$

$$\text{Best} = B = (1.25, 0.75)$$

$$\text{Good} = G = (1.5, 0)$$

$$\text{Worst} = W = (2, 0.5)$$



Midpoint of Good and Best

$$M = \frac{B+G}{2} = \left(\frac{1.25+1.5}{2}, \frac{0.75+0}{2} \right)$$

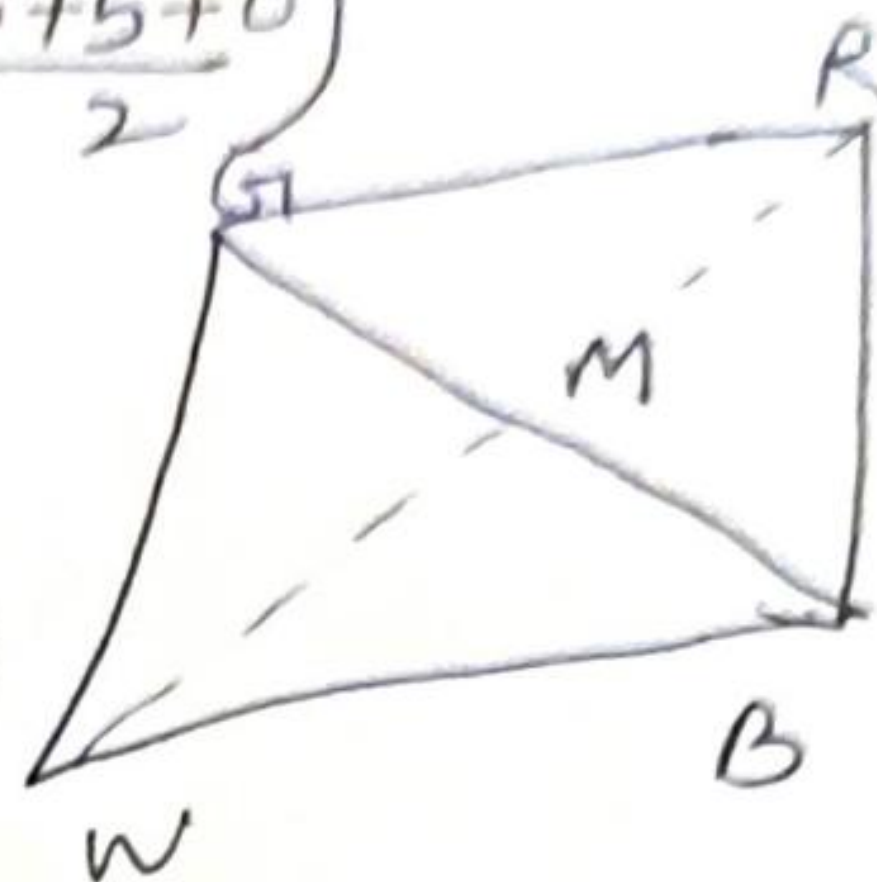
$$M = (1.375, 0.375)$$

Now,

$$R = 2M - W$$

$$= 2(1.375, 0.375) - (2, 0.5)$$

$$R = (0.75, 0.25)$$



$$f(R) = 1.560$$

$$\Rightarrow f(R) < f(G) \quad \text{i.e.} \quad 1.560 < 3.276$$

Begins Case (i)

$$\Rightarrow f(B) \neq f(R) \quad \text{i.e.} \quad 2.194 \neq 1.560$$

Compute E with $f(E)$

$$E = 2R - M = 2(0.75, 0.25) - (1.375, 0.375)$$

$$E = (0.125, 0.125)$$

$$f(E) = 1.016$$

$$f(E) < f(B) \quad \text{i.e.} \quad 1.016 < 2.194$$

Replace W with E.

Iteration III!

find BGW triangle

Now Vertices are $(1.25, 0.75)$, $(1.5, 0)$
 $(0.125, 0.125)$

$$f(1.25, 0.75) = 2.194$$

$$f(1.5, 0) = 3.276$$

$$f(0.125, 0.125) = 1.016$$

$$\text{Best} = B = (0.125, 0.125)$$

$$\text{Good} = G = (1.25, 0.75)$$

$$\text{Worst} = W = (1.5, 0)$$

Mid point of Good and Best

$$M = \frac{B+G}{2}$$

$$M = \left(\frac{0.125 + 1.25}{2}, \frac{0.125 + 0.75}{2} \right)$$

$$M = (0.6875, 0.4375)$$

$$\text{Now, } R = 2M - W$$

$$R = 2(0.6875, 0.4375) - (1.5, 0)$$

$$R = (-0.125, 0.875)$$

$$R = 0.402$$

$$\Rightarrow f(R) < f(G) \quad \text{i.e. } 0.402 < 2.194$$

Begins case (i)

$$\Rightarrow f(B) \neq f(R) \quad \text{i.e. } 1.016 < 0.402$$

Compute E and $f(E)$

$$E = 2R - M$$

$$= 2(-0.125, 0.875) - (0.6875, 0.4375)$$

$$E = (-0.9375, 1.3125)$$

$$f(E) = 2.453$$

$$\Rightarrow f(E) \neq f(B) \quad \text{i.e. } 2.453 \neq 1.016$$

Replace with R.
and no an.

